



```
In [22]: import pandas as pd
import torch
import torch.nn as nn
import numpy as np
import matplotlib.pyplot as plt
from torch.utils.data import Dataset, DataLoader
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from torch.nn.utils.rnn import pad_sequence
import warnings
warnings.filterwarnings("ignore", category=DeprecationWarning)
```

```
In [23]: # Device configuration
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")
```

Using device: cuda

```
In [24]: # Load and prepare data
data = pd.read_csv("/content/ner_dataset.csv", encoding="latin1").ffill()
words = list(data["Word"].unique())
tags = list(data["Tag"].unique())

if "ENDPAD" not in words:
    words.append("ENDPAD")

word2idx = {w: i + 1 for i, w in enumerate(words)}
tag2idx = {t: i for i, t in enumerate(tags)}
idx2tag = {i: t for t, i in tag2idx.items()}
```

```
In [25]: data.head(50)
```

Out[25]:

	Sentence #	Word	POS	Tag
0	Sentence: 1	Thousands	NNS	O
1	Sentence: 1	of	IN	O
2	Sentence: 1	demonstrators	NNS	O
3	Sentence: 1	have	VBP	O
4	Sentence: 1	marched	VBN	O
5	Sentence: 1	through	IN	O
6	Sentence: 1	London	NNP	B-geo
7	Sentence: 1	to	TO	O
8	Sentence: 1	protest	VB	O
9	Sentence: 1	the	DT	O
10	Sentence: 1	war	NN	O
11	Sentence: 1	in	IN	O
12	Sentence: 1	Iraq	NNP	B-geo
13	Sentence: 1	and	CC	O
14	Sentence: 1	demand	VB	O
15	Sentence: 1	the	DT	O
16	Sentence: 1	withdrawal	NN	O
17	Sentence: 1	of	IN	O
18	Sentence: 1	British	JJ	B-gpe
19	Sentence: 1	troops	NNS	O
20	Sentence: 1	from	IN	O
21	Sentence: 1	that	DT	O
22	Sentence: 1	country	NN	O
23	Sentence: 1	.	.	O
24	Sentence: 2	Families	NNS	O
25	Sentence: 2	of	IN	O
26	Sentence: 2	soldiers	NNS	O
27	Sentence: 2	killed	VBN	O
28	Sentence: 2	in	IN	O
29	Sentence: 2	the	DT	O
30	Sentence: 2	conflict	NN	O

	Sentence #	Word	POS	Tag
31	Sentence: 2	joined	VBD	O
32	Sentence: 2	the	DT	O
33	Sentence: 2	protesters	NNS	O
34	Sentence: 2	who	WP	O
35	Sentence: 2	carried	VBD	O
36	Sentence: 2	banners	NNS	O
37	Sentence: 2	with	IN	O
38	Sentence: 2	such	JJ	O
39	Sentence: 2	slogans	NNS	O
40	Sentence: 2	as	IN	O
41	Sentence: 2	"	``	O
42	Sentence: 2	Bush	NNP	B-per
43	Sentence: 2	Number	NN	O
44	Sentence: 2	One	CD	O
45	Sentence: 2	Terrorist	NN	O
46	Sentence: 2	"	``	O
47	Sentence: 2	and	CC	O
48	Sentence: 2	"	``	O
49	Sentence: 2	Stop	VB	O

Essential info about tagged entities:

```

geo = Geographical Entity
org = Organization
per = Person
gpe = Geopolitical Entity
tim = Time indicator
art = Artifact
eve = Event
nat = Natural Phenomenon

```

```

In [26]: print("Unique words in corpus:", data['Word'].nunique())
print("Unique tags in corpus:", data['Tag'].nunique())

```

```

Unique words in corpus: 35177
Unique tags in corpus: 17

```

```

In [27]: print("Unique tags are:", tags)

```

Unique tags are: ['O', 'B-geo', 'B-gpe', 'B-per', 'I-geo', 'B-org', 'I-org', 'B-tim', 'B-art', 'I-art', 'I-per', 'I-gpe', 'I-tim', 'B-nat', 'B-eve', 'I-eve', 'I-nat']

```
In [28]: # Group words by sentences
class SentenceGetter:
    def __init__(self, data):
        self.grouped = data.groupby("Sentence #", group_keys=False).apply(
            lambda s: [(w, t) for w, t in zip(s["Word"], s["Tag"])]
        )
        self.sentences = list(self.grouped)

getter = SentenceGetter(data)
sentences = getter.sentences
```

```
In [29]: sentences[35]
```

```
Out[29]: [('The', 'O'),
          ('U.S.', 'B-org'),
          ('Geological', 'I-org'),
          ('Survey', 'I-org'),
          ('gave', 'O'),
          ('a', 'O'),
          ('preliminary', 'O'),
          ('estimate', 'O'),
          ('of', 'O'),
          ('the', 'O'),
          ('strength', 'O'),
          ('of', 'O'),
          ('the', 'O'),
          ('Tuesday', 'B-tim'),
          ('morning', 'I-tim'),
          ('quake', 'O'),
          ('at', 'O'),
          ('6.7', 'O'),
          ('on', 'O'),
          ('the', 'O'),
          ('Richter', 'B-geo'),
          ('scale', 'O'),
          (',', 'O'),
          ('and', 'O'),
          ('said', 'O'),
          ('the', 'O'),
          ('epicenter', 'O'),
          ('was', 'O'),
          ('close', 'O'),
          ('to', 'O'),
          ('the', 'O'),
          ('island', 'O'),
          ('of', 'O'),
          ('Nias', 'B-org'),
          ('.', 'O')]
```

```
In [30]: # Encode sentences
X = [[word2idx[w] for w, t in s] for s in sentences]
y = [[tag2idx[t] for w, t in s] for s in sentences]
```

```
In [31]: word2idx
```

```
Out[31]: {'Thousands': 1,
          'of': 2,
          'demonstrators': 3,
          'have': 4,
          'marched': 5,
          'through': 6,
          'London': 7,
          'to': 8,
          'protest': 9,
          'the': 10,
          'war': 11,
          'in': 12,
          'Iraq': 13,
          'and': 14,
          'demand': 15,
          'withdrawal': 16,
          'British': 17,
          'troops': 18,
          'from': 19,
          'that': 20,
          'country': 21,
          '.': 22,
          'Families': 23,
          'soldiers': 24,
          'killed': 25,
          'conflict': 26,
          'joined': 27,
          'protesters': 28,
          'who': 29,
          'carried': 30,
          'banners': 31,
          'with': 32,
          'such': 33,
          'slogans': 34,
          'as': 35,
          '"': 36,
          'Bush': 37,
          'Number': 38,
          'One': 39,
          'Terrorist': 40,
          'Stop': 41,
          'Bombings': 42,
          'They': 43,
          'Houses': 44,
          'Parliament': 45,
          'a': 46,
          'rally': 47,
          'Hyde': 48,
          'Park': 49,
          'Police': 50,
          'put': 51,
          'number': 52,
          'marchers': 53,
          'at': 54,
```

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'it': 59,
'was': 60,
'1,00,000': 61,
'The': 62,
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'on': 64,
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'conference': 67,
'Britain': 68,
's': 69,
'ruling': 70,
'Labor': 71,
'Party': 72,
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'English': 74,
'seaside': 75,
'resort': 76,
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'party': 78,
'is': 79,
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'Atomic': 100,
'Energy': 101,
'Agency': 102,
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'could': 159,
'impose': 160,
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'finds': 163,
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'Nuclear': 166,
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'new': 169,
'President': 170,
'Mahmoud': 171,
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'fuel': 179,
'program': 180,
'are': 181,
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'nation': 183,
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'Germans': 185,
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'Nigerian': 187,
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'were': 190,
'kidnapped': 191,
'by': 192,
'armed': 193,
'militants': 194,
'during': 195,
'raid': 196,
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'Nigeria': 198,
'oil-rich': 199,
'Delta': 200,
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'Berger': 207,
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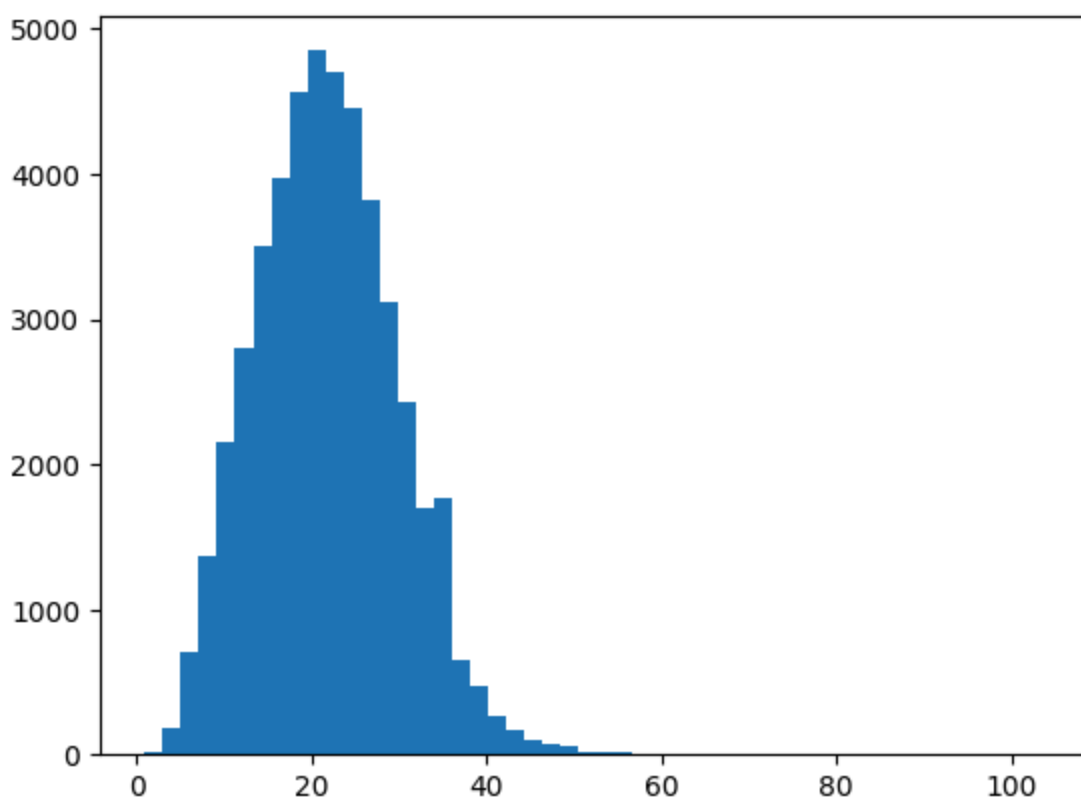
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'spared': 998,  
'bloodshed': 999,  
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...}
```

```
In [32]: plt.hist([len(s) for s in sentences], bins=50)  
plt.show()
```

```
In [33]: # Pad sequences
max_len = 50
X_pad = pad_sequence([torch.tensor(seq) for seq in X], batch_first=True, padding_value=0)
y_pad = pad_sequence([torch.tensor(seq) for seq in y], batch_first=True, padding_value=0)
X_pad = X_pad[:, :max_len]
y_pad = y_pad[:, :max_len]
```

```
In [34]: X_pad[0]
```

[illegible]

```
In [35]: y_pad[0]
```

```
Out[35]: tensor([0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0,  
                0,  
                0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,  
                0,  
                0, 0])
```

```
In [36]: # Train/test split
X_train, X_test, y_train, y_test = train_test_split(X_pad, y_pad, test_size=0.
```

```
In [37]: # Dataset class
```

```

class NERDataset(Dataset):
    def __init__(self, X, y):
        self.X = X
        self.y = y

    def __len__(self):
        return len(self.X)

    def __getitem__(self, idx):
        return {
            "input_ids": self.X[idx],
            "labels": self.y[idx]
        }

train_loader = DataLoader(NERDataset(X_train, y_train), batch_size=32, shuffle=True)
test_loader = DataLoader(NERDataset(X_test, y_test), batch_size=32)

```

```

In [38]: class BiLSTMTagger(nn.Module):
    def __init__(self, vocab_size, tagset_size, embedding_dim=50, hidden_dim=100):
        super(BiLSTMTagger, self).__init__()
        self.embedding = nn.Embedding(vocab_size, embedding_dim, padding_idx=0)
        self.dropout = nn.Dropout(dropout)
        self.lstm = nn.LSTM(embedding_dim, hidden_dim, batch_first=True, bidirectional=True)
        self.fc = nn.Linear(hidden_dim * 2, tagset_size)

    def forward(self, x):
        x = self.embedding(x)
        x = self.dropout(x)
        x, _ = self.lstm(x)
        out = self.fc(x)
        return out

vocab_size = len(word2idx) + 1
tagset_size = len(tag2idx)

```

```

In [39]: device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = BiLSTMTagger(len(word2idx) + 1, len(tag2idx)).to(device)
loss_fn = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)

```

```

In [40]: def train_model(model, train_loader, test_loader, loss_fn, optimizer, epochs=30):
    train_losses, val_losses = [], []
    for epoch in range(epochs):
        model.train()
        total_loss = 0
        for batch in train_loader:
            input_ids = batch["input_ids"].to(device)
            labels = batch["labels"].to(device)
            optimizer.zero_grad()
            outputs = model(input_ids)
            loss = loss_fn(outputs.view(-1, len(tag2idx)), labels.view(-1))
            loss.backward()

```

```

        optimizer.step()
        total_loss += loss.item()
    train_losses.append(total_loss)

    model.eval()
    val_loss = 0
    with torch.no_grad():
        for batch in test_loader:
            input_ids = batch["input_ids"].to(device)
            labels = batch["labels"].to(device)
            outputs = model(input_ids)
            loss = loss_fn(outputs.view(-1, len(tag2idx)), labels.view(-1))
            val_loss += loss.item()
    val_losses.append(val_loss)
    print(f"Epoch {epoch+1}: Train Loss = {total_loss:.4f}, Val Loss = {val_loss:.4f}")
    return train_losses, val_losses

```

In [41]:

```

def evaluate_model(model, test_loader, X_test, y_test):
    model.eval()
    true_tags, pred_tags = [], []
    with torch.no_grad():
        for batch in test_loader:
            input_ids = batch["input_ids"].to(device)
            labels = batch["labels"].to(device)
            outputs = model(input_ids)
            preds = torch.argmax(outputs, dim=-1)
            for i in range(len(labels)):
                for j in range(len(labels[i])):
                    if labels[i][j] != tag2idx["0"]:
                        true_tags.append(idx2tag[labels[i][j].item()])
                        pred_tags.append(idx2tag[preds[i][j].item()])

```

In [42]:

```

train_losses, val_losses = train_model( model, train_loader, test_loader, loss_fn,
                                         evaluate_model( model, test_loader, X_test, y_test ) )

```

Epoch 1: Train Loss = 257.5033, Val Loss = 31.6766

Epoch 2: Train Loss = 106.7530, Val Loss = 22.5612

Epoch 3: Train Loss = 79.5082, Val Loss = 19.1275

In [43]:

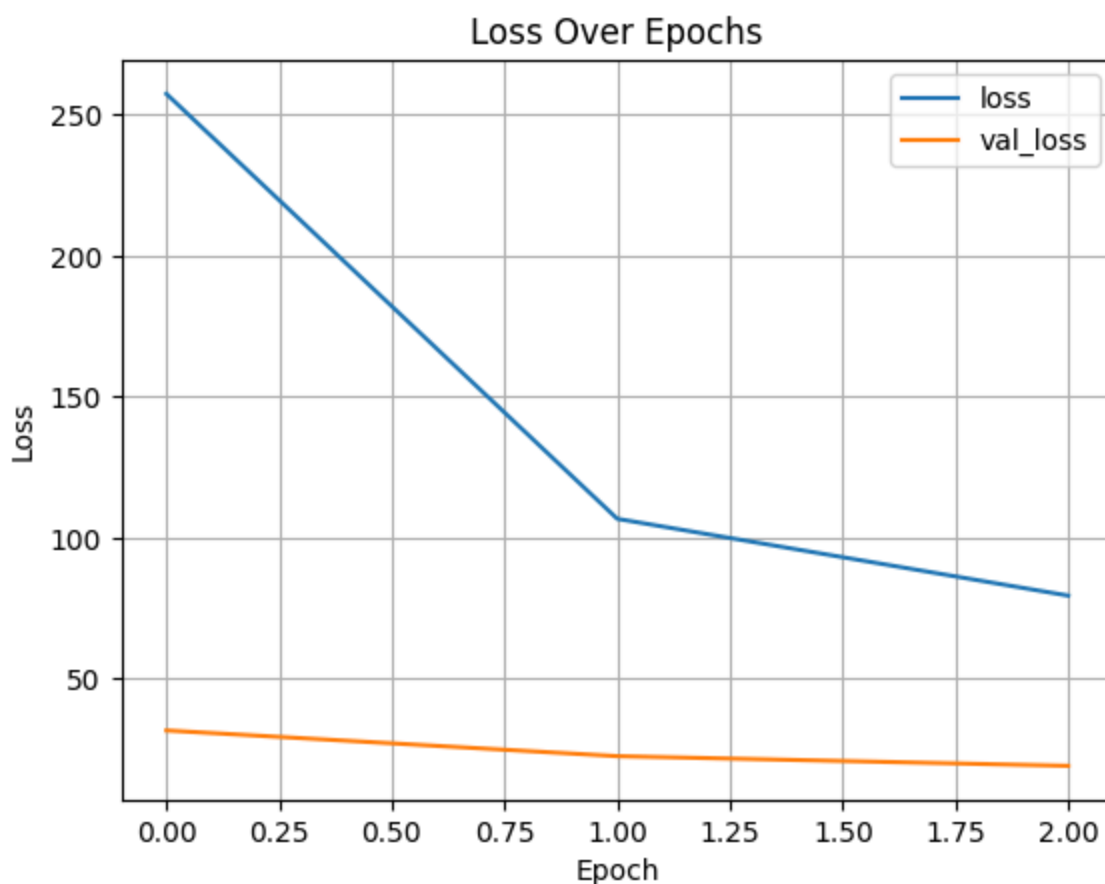
```

# Plot loss
print('Name:      Y.NITHEESH      ')
print('Register Number: 212224040370  ')
history_df = pd.DataFrame({"loss": train_losses, "val_loss": val_losses})
history_df.plot(title="Loss Over Epochs")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.grid(True)
plt.show()

```

Name: Y.NITHEESH

Register Number: 212224040370



```
In [45]: # Inference and prediction
i = 125
model.eval()
sample = X_test[i].unsqueeze(0).to(device)
output = model(sample)
preds = torch.argmax(output, dim=-1).squeeze().cpu().numpy()
true = y_test[i].numpy()

print('Name:      Y.NITHEESH          ')
print('Register Number: 212224040370  ')
print("{:<15} {:<10} {}\\n{}".format("Word", "True", "Pred", "-" * 40))
for w_id, true_tag, pred_tag in zip(X_test[i], y_test[i], preds):
    if w_id.item() != word2idx["ENDPAD"]:
        word = words[w_id.item() - 1]
        true_label = tags[true_tag.item()]
        pred_label = tags[pred_tag]
        print(f"{word:<15} {true_label:<10} {pred_label}")
```

Name: Y.NITHEESH

Register Number: 212224040370

Word	True	Pred

Palestinian	B-gpe	B-gpe
officials	0	0
say	0	0
two	0	0
Palestinians	B-gpe	B-gpe
have	0	0
been	0	0
killed	0	0
in	0	0
an	0	0
accidental	0	0
explosion	0	0
in	0	0
a	0	0
West	B-org	B-org
Bank	I-org	I-org
refugee	0	0
camp	0	0
.	0	0